INTRODUCTION TO IP ADDRESSING THEORY

1. List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Difference Between IPv4 and IPv6:-

IPv4	IPv6
1. IPv4 uses a **32-bit** address.	1. IPv6 uses a **128-bit** address.
2. It is written in decimal format.	2. It is written in hexadecimal format.
3. Example: 192.168.1.1	3. Example: 2001:db8::1
4. It supports approximately **4.3 billion** unique addresses.	4. It supports a much larger number of unique addresses.
5. NAT (Network Address Translation) is commonly used.	5. NAT is generally not required because of its huge address space.
6. Security features such as IPsec can be used.	6. IPsec support is standardized, but it is not automatically enabled in every connection.
7.. IPv4 headers are more complex for basic packet processing.	7.. IPv6 has a simpler base header for more efficient packet processing.

2. Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Static IPv4 Address :-

Network: 192.168.10.0/24
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.10.1

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.11	255.255.255.0	192.168.10.1
Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
Printer	192.168.10.13	255.255.255.0	192.168.10.1

Explanation:-

1. **Unique IP Address:** Each device has a different IP address to avoid IP conflicts.
2. **Subnet Mask:** The subnet mask 255.255.255.0 identifies the network and host portions of each IP address.
3. **Default Gateway:** The address 192.168.10.1 represents the router, which forwards traffic to other networks, such as the internet.
4. **Network Address:** 192.168.10.0 identifies the network, so it is not assigned to an individual device.

3. Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

- Network: 192.168.1.0/24
- CIDR: /27
- Subnet Mask: 255.255.255.224

Calculation:

- Number of subnets: 2^3 = **8**
- Usable hosts per subnet: $2^5 - 2$ = **30**
- Block size: 32

Subnet Ranges:

Subnet	Network Address	Usable Host IPs	Broadcast Address
1	192.168.1.0	192.168.1.1–30	192.168.1.31
2	192.168.1.32	192.168.1.33–62	192.168.1.63
3	192.168.1.64	192.168.1.65–94	192.168.1.95

4. You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

Given:

- IP Address: 10.0.5.17
- Subnet Mask: 255.255.255.0
- CIDR Notation: /24

Answer:

1. **Network Address:** 10.0.5.0
2. **Broadcast Address:** 10.0.5.255
3. **Valid Host Range:** 10.0.5.1 – 10.0.5.254
4. **Total Usable Hosts:** 254

5. Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

